

BAYESIAN PREDICTION OF HIV PREVALENCE

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Summary: For the last six years, the Department of Health in South Africa has monitored the prevalence of the HIV virus in attenders of antenatal clinics. Using the model of exponential growth, this data has been used to make short-term predictions of future prevalence rates. The exponential model has fitted the data well so far, but the lengthening of doubling times over the years suggests that this model would not be adequate over a longer period. The Michaelis-Menten model for prevalence data follows exponential growth in the initial stages, but allows for an upper bound. Modelling the prevalence at time t as a Beta variable with expectation equal to the M-M model, and with noninformative priors, the predictive densities of expected prevalence at some future times were obtained using the Gibbs sampler. The marginal posterior distribution of the parameter representing maximum prevalence was also obtained. This was done for the whole country as well as for a higher risk subgroup.

1. Introduction

The Department of National Health and Population Development runs a programme for the prevention of haemolytic disease of the newborn. All women attending antenatal clinics are screened by having a blood specimen sent to one of several laboratories for, inter alia, Rh and ABO testing.

During 1990, the Department undertook a national survey to determine the extent of HIV infection in the country. Because of the logistical problems that would be encountered if members of the general population were to be included in the survey, it was decided to focus on women attending antenatal clinics.

The number of Rh/ABO tests paid for by the Department was used as a basis for planning the sampling strategy. According to Epidemiological Comments (1991), 538911 such tests were carried out during 1990, covering all population groups. Blood specimens received during October and November 1990 were sampled and anonymously tested for HIV antibodies. The sample was stratified by population group and each positive result was associated with population group, age and place of residence of the women. The total sample size for 1990 was 14376 and the estimated HIV prevalence 0.68%.

The sampling procedure was then repeated for the following years (1991–1995). Based on the number of blood specimens processed by the participating laboratories, a time interval was calculated during which the required number of specimens would be obtained. These time intervals varied by population group, namely Blacks – one week, Coloureds – four weeks, whites – four weeks and Asians – eight weeks. Laboratories were requested to collect consecutive blood specimens for these time intervals during October and November. The number of women attending the antenatal clinics yearly were more than half a million and the sample sizes over the six years varied between 14000 and 19000. The estimates for the whole country are weighted averages over population groups and provinces.

There is a possibility that during the time period 1990–1995 some of the women became pregnant more than once or even every year, which means that

they attended the antenatal clinics quite often. It is, however, very unlikely that they or any other women would have been sampled twice. The reason for this is that only one specific week during October and November was used to obtain the required number of specimens for the "high pregnancy group".

From the sampling method it therefore seems quite reasonable to assume that the observed prevalence p_t ($t = 1, \dots, N$), where $N = 6$ (the number of years), are independently distributed. The results were published in *Epidemiological Comments* (1991–1996), a monthly publication of the Directorate: Epidemiology of the Department of Health and Population Development.

In this paper we will analyse only the data for the whole country and the province of KwaZulu-Natal, a high risk region. The yearly estimates are given in table 1.

Table 1: HIV prevalence in women attending antenatal clinics (%)

	1990	1991	1992	1993	1994	1995
South Africa	0.68	1.35	2.42	4.25	7.57	10.44
KwaZulu-Natal	1.61	2.87	4.77	9.26	14.35	18.23

Using the model of exponential growth,

$$p_t = \gamma e^{\lambda t},$$

the data has been used to extrapolate and make short-term predictions of future prevalence rates. This model has fitted the data very well for the first five years when prevalence is low, it is simple and easy to fit. The doubling time is constant and equal to $\frac{\ln 2}{\lambda}$. However, it can obviously not be true in the

long term and would typically overestimate future prevalence. Department of Health estimates of the doubling time, 12.05 months in 1993 and 13.07 months in 1994, already indicate a lengthening of the doubling time, and this trend should continue as data accumulates.

2. Michaelis-Menten Model

As the exponential model is unrealistic later in the epidemic, we will consider the Michaelis-Menten model, allowing for estimation of maximum future prevalence. The relation between observed prevalence p_t and time t can be written as

$$p_t = \frac{\gamma e^{\lambda t}}{\theta + e^{\lambda t}} \quad (2.1)$$

This model has been applied in a number of biological and biochemical situations, particularly in the study of enzyme kinetics. See, for example, Beverton and Holt (1957), Bartholomay (1962), Dowd and Riggs (1965) and Cornish-Bowden (1981).

Normally the parameter space for γ is unbounded to the right. In our application where the response variable is a proportion, γ represents maximum prevalence and is bounded in the interval $(0,1]$. The other two parameters, θ and λ , are assumed to have positive support. This model follows exponential growth in the early stages and the initial doubling time is the same as that of the exponential model, namely $\frac{\ln 2}{\lambda}$. In general, if $\theta \rightarrow \infty$ and $\frac{\gamma}{\theta} \rightarrow \alpha$, the Michaelis-Menten model approaches the exponential model if γ is unbounded to the right.

The doubling time from time t is given by

$$\tau_t = \frac{1}{\lambda} \ln \left(\frac{2\theta}{\theta - e^{\lambda t}} \right) \quad \text{for } t < \frac{1}{\lambda} \ln \theta, \quad (2.2)$$

and the time to $100\beta\%$ of maximum prevalence is

$$\delta_\beta = \frac{1}{\lambda} \ln \left(\frac{\theta\beta}{1 - \beta} \right). \quad (2.3)$$

There are different views on how best to estimate the parameters in (2.1), and a considerable number of papers have appeared on the subject, see for example, Eienthal and Cornish-Bowden (1974), Atkins and Nimmo (1975), Tong and Metzler (1980), Currie (1982), Ruppert, Cressie and Carrol (1989) and Nelder (1991). One reason for the differing views on estimation is the uncertainty about the error structure of the data, and the fitting method depends on the subject area, experiment and underlying assumptions. Another reason is the high breakdown nature of the Michaelis-Menten model (Stromberg, 1993), leading to a search for more robust estimators. The most commonly used fitting method seems to be linearization such as the double reciprocal transformation of Lineweaver and Burk (1934), or the Woolf transformation (Haldane, 1957).

3. A Bayesian model for prevalence data

We assume that the observed prevalence at time t has a Beta distribution with expectation equal to the Michaelis-Menten model in (2.1). So $p_t \sim \text{Beta}(\alpha, \beta_t)$ with

$$E[p_t | \gamma, \theta, \lambda] = \frac{\alpha}{\alpha + \beta_t} = \frac{\gamma e^{\lambda t}}{\theta + e^{\lambda t}} = \mu_t. \quad (3.1)$$

It then follows that

$$\text{Var}[p_t|\gamma, \theta, \lambda, \alpha] = \frac{\mu_t^2(1 - \mu_t)}{\alpha + \mu_t} \quad (3.2)$$

and β_t is uniquely determined as

$$\beta_t = \alpha f_t, \quad \text{where} \quad f_t = \frac{1 - \mu_t}{\mu_t} \quad (3.3)$$

This means that if we assume additive errors, $p_t = \mu_t + \varepsilon_t$, then

$$f(\varepsilon_t|\mu_t, \alpha) = \frac{1}{B(\alpha, \beta_t)} (\mu_t + \varepsilon_t)^{\alpha-1} (1 - \mu_t - \varepsilon_t)^{\beta_t-1}, \quad -\mu_t < \varepsilon_t < 1 - \mu_t, \quad (3.4)$$

conditionally independent of each other, with mean zero and variance as in (3.2). For fixed μ_t the error variance is a decreasing function of α , and for fixed $0 < \alpha < \infty$, maximum error variance occurs when $\frac{1}{2} < \mu_t < \frac{2}{3}$.

The likelihood function is then given by

$$L(\gamma, \theta, \lambda, \alpha|p_t, t = 1, \dots, N) = \prod_{t=1}^N \frac{1}{B(\alpha, \alpha f_t)} p_t^{\alpha-1} (1 - p_t)^{\alpha f_t-1}. \quad (3.5)$$

The prior distribution of the four parameters is assumed to be of the form

$$\pi(\gamma, \theta, \lambda, \alpha) \propto \pi(\alpha|\gamma, \theta, \lambda) \pi(\gamma) \pi(\theta) \pi(\lambda) \quad (3.6)$$

where $\pi(\gamma)$, $\pi(\theta)$ and $\pi(\lambda)$ are locally uniform and $\pi(\alpha|\gamma, \theta, \lambda)$ is a conditional Jeffreys prior,

$$\pi(\alpha|\gamma, \theta, \lambda) \propto I_\alpha^{-\frac{1}{2}} \quad (3.7)$$

where I_α is the Fisher information,

$$I_\alpha = -E \left[\frac{\partial^2 L}{\partial \alpha^2} \right]. \quad (3.8)$$

We have considered the full joint Jeffreys prior or reference prior for all parameters, but the determinant of the Fisher information matrix is extremely

messy. Another option was to use the conditional Jeffreys priors for each parameter consecutively in the Gibbs sampling process, but these priors had virtually no effect on the likelihood function, except in the case of the scale parameter α .

From (3.5) and (3.8) it follows that

$$I_\alpha = N\psi'(\alpha) + \sum_{t=1}^N [f_t^2\psi'(\alpha f_t) - (1 + f_t)^2\psi'(\alpha + \alpha f_t)] \quad (3.9)$$

where f_t is given in (3.3) and $\psi'(\cdot)$ is the trigamma function,

$$\psi'(\alpha) = \frac{d^2}{d\alpha^2} \ell n \Gamma(\alpha) = \sum_{k=0}^{\infty} \frac{1}{(\alpha + k)^2}. \quad (3.10)$$

For n an integer, a simple formula is

$$\psi'(n) = \frac{\pi^2}{6} - \sum_{k=1}^{n-1} \frac{1}{k^2}.$$

So the full conditional distributions of γ, θ and λ are proportional to the likelihood in (3.5), while

$$\pi(\alpha|p_t, f_t) \propto L(\alpha, \gamma, \theta, \lambda|p_t) \times I_\alpha^{\frac{1}{2}}. \quad (3.11)$$

4. Results

The marginal posterior distribution of the parameters were obtained by means of the Gibbs sampler (Geman and Geman, 1984, Gelfand and Smith, 1990). Observations were generated successively from the conditional distributions by means of a Bayesian bootstrap where the distributions are discretized. Convergence to the marginal posteriors of α and λ is fast, but quite slow for θ and γ , mainly because of strong correlation between θ and γ . This

correlation is 0.953 for the KwaZulu-Natal data and 0.987 for the whole country. This is an indication that the data fits an exponential model quite well, in which case correlation between γ and θ would be perfect. Simulation is slow due to the time used in evaluating (3.5) and (3.9). For the South African data, for example, α is in the range (0 – 2500) and when expected prevalence, μ_t , is low, $\beta_t = \alpha f_t$ can be several hundred thousands, making accurate evaluations quite lengthy.

The simulations were started with the least squares estimators and after a burn-in period of ± 2000 , 50000 observations were generated from the conditional distributions. The marginal posteriors were obtained by averaging over the conditionals.

In the case of the functions (2.2) and (2.3) the histograms were used to obtain posteriors.

Figure 1 shows the marginal posterior distribution of maximum prevalence, γ , for the whole country and for the province of KwaZulu-Natal. The summary statistics of all parameters are given in table 1.

Figure 2 shows expected times to 90% of maximum prevalence, and figure 3 shows expected doubling time as from 1995, given that the current prevalence rate will double. The probabilities for doubling are 0.661 and 0.193 for the country and KwaZulu-Natal respectively.

Figure 4 is the expected prevalences for 1996 and figures 5 and 6 show posterior expectations up to the year 2005 together with the posterior modes and a 95% credibility interval.

Table 1: Posterior Measures*

	Parameter	Mean	Median	Mode	Variance	95% CI	LSE
SA	γ	0.2805	0.2400	0.2080	0.0166	0.160 – 0.670	0.1680
	θ	75.0408	66.0000	60.078	870.897	47.3 – 164.2	57.942
	λ	0.6660	0.6663	0.6717	0.0018	0.584 – 0.748	0.7620
	α	407	279	244	98404	39 – 1320	–
	δ_{90}	3.7624	3.5670	3.0873	1.1090	2.207 – 6.379	2.211
	τ_{95}	3.1181	2.7480	1.7591	2.0237	1.536 – 7.092	–
	μ_{96}	0.1508	0.1484	0.1437	0.00028	0.125 – 0.188	0.1313
KZN	γ	0.3483	0.3240	0.3032	0.0084	0.244 – 0.616	0.2440
	θ	41.1093	39.031	37.220	73.465	31.23 – 66.41	38.039
	λ	0.6696	0.6720	0.6755	0.0020	0.578 – 0.754	0.7923
	α	192	167	163	16118	26 – 509	–
	δ_{90}	2.8561	2.7170	2.4235	0.6866	1.578 – 4.932	1.366
	τ_{95}	4.0386	3.6080	2.2953	2.9863	2.071 – 8.743	–
	μ_{96}	0.2427	0.2398	0.2345	0.00049	0.206 – 0.296	0.2125

* δ_{90} = time to 90% of maximum prevalence (from 1995)

τ_{95} = doubling time from 1995

μ_{96} = expected prevalence for 1996.

The LS estimators in the table were obtained by simply finding the set of parameters from the Gibbs sampling which minimizes the square error, and the ML estimators were obtained by maximizing the likelihood given in (3.5).

The results are given in table 1 and show closer agreement with posterior modes than with posterior means.

5. Conclusions

This study was originally done with the data from the first five years, but the posterior distributions of the parameters, especially γ , indicated that there was very little information in the data concerning maximum prevalence, and could not really discriminate between the exponential and Michaelis-Menten models. The 1995 figures, however, changed that and had a marked influence on the estimations.

For estimating maximum prevalence, the posterior modes seem to be much more realistic figures than the posterior means, 0.208 against 0.281 for the country and 0.303 against 0.348 for KwaZulu-Natal. The posterior distributions of γ are still very diffused and a few more years of data should reduce the uncertainty and bring the means closer to the modes.

In this study sample sizes were ignored and the observed prevalence rates treated as Beta variables. Another approach would be to treat the observed positive cases at time t as a Binomial variable, $y_t \sim \text{Bin}(\mu_t, n_t)$, where n_t is sample size and μ_t given in (3.1). This model has only three parameters, with $\text{Var}[p_t|\gamma, \theta, \lambda] = \frac{\mu_t(1-\mu_t)}{n_t}$, so it loses some flexibility. Also, the assumption of constant μ_t over the population from which the sample is drawn, must be justifiable. This model will be considered in the future.

Note:

Since the analysis was done the figures for 1996 became available (Epidemiological Comment, 1997). The national estimate is 0.1417 and the KwaZulu-Natal estimate is 0.1990. Comparing these values with the modes of μ_{96} in table 1, 0.1437 and 0.2345 respectively, we see that the national figure is quite close to the posterior mode. The KwaZulu-Natal figure, however, is surprisingly low. It is in fact just outside the 95% credibility interval. It seems that there is a sharp drop in the rate of increase for this region, which can affect the estimated maximum prevalence considerably.

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7. References

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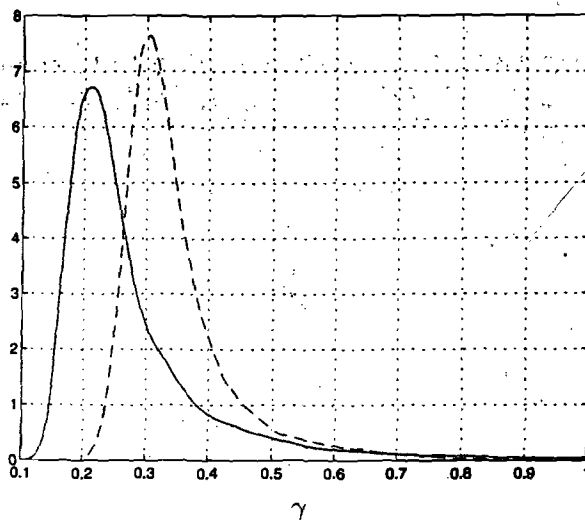


Figure 1. Posterior distribution of maximum prevalence γ for South Africa (—) and KwaZulu-Natal (---)

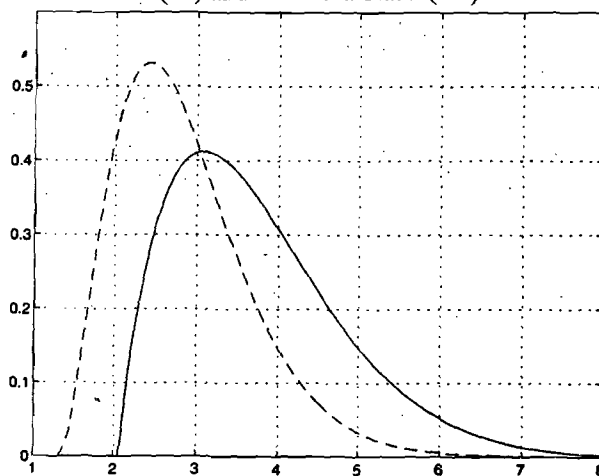


Figure 2. Posterior distribution of time to 90% of maximum prevalence δ_{90} for South Africa (—) and KwaZulu-Natal (---)

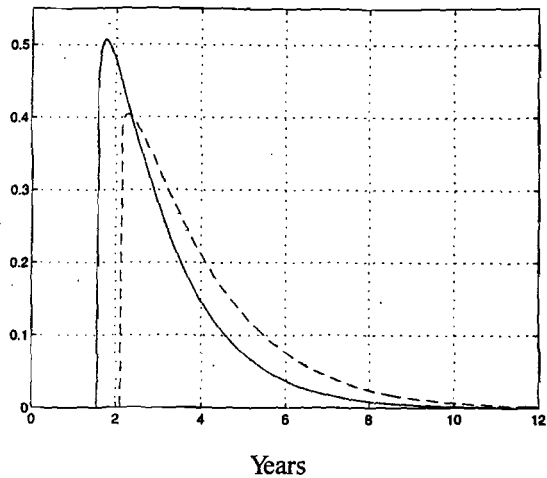


Figure 3. Posterior distribution of doubling time from 1995, τ_{95} for South Africa (—) and KwaZulu-Natal (---)

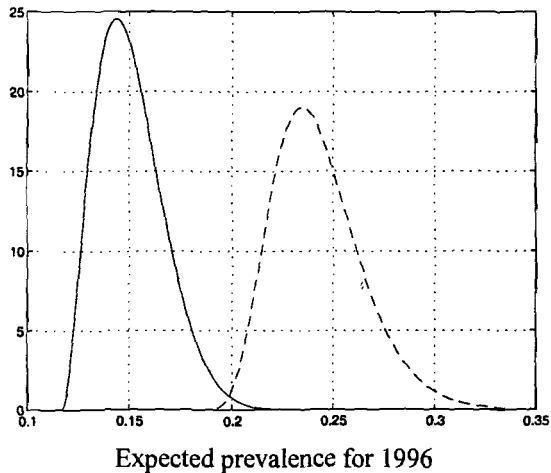


Figure 4. Posterior distribution of expected prevalence for 1996, μ_{96} for South Africa (—) and KwaZulu-Natal (---)

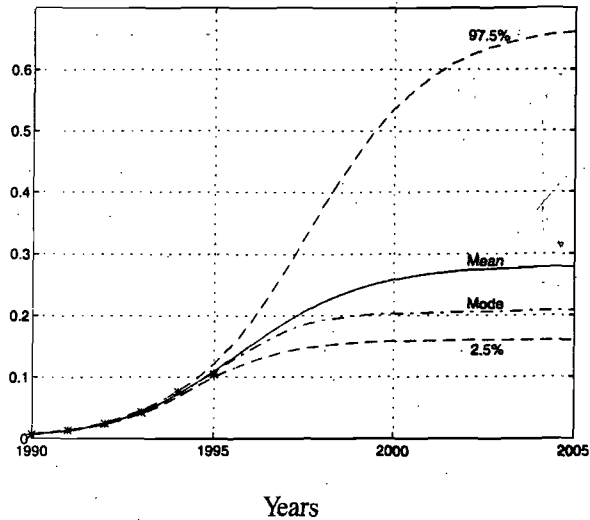


Figure 5. Observed and expected prevalences to 2005 for South Africa

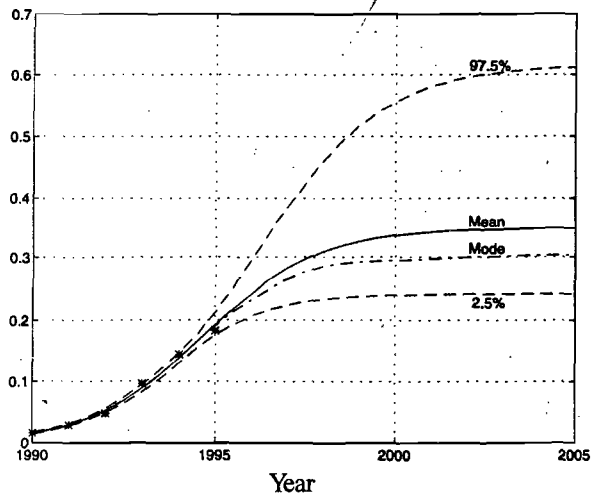


Figure 6. Observed and expected prevalences to 2005 for KwaZulu-Natal